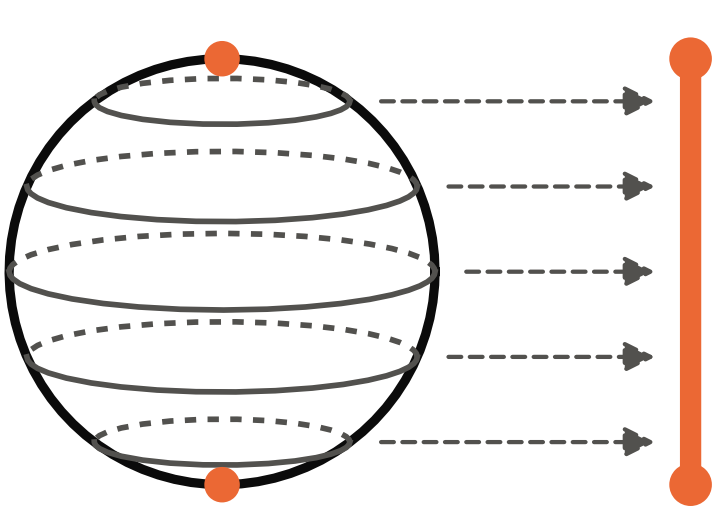
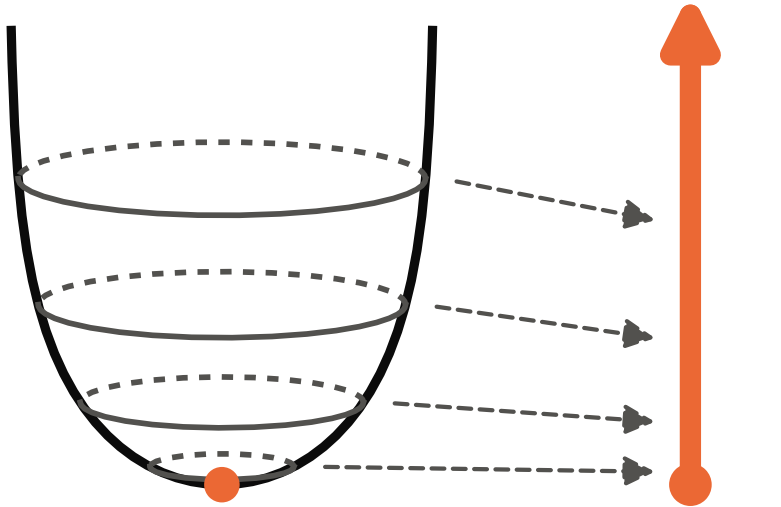

S^2
$[-1, 1]$


cigar
$[0, \infty)$